



Jaehyun Byun

ID KAIST

Design Researcher

#Human-Computer Interaction #Prototyping
#Extended Reality #AI Experience

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EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** 2026.08 -
Master of Engineering, Industrial Design (Advisor: Andrea Bianchi)
- **Kyung Hee University** 2020.03 - 2026.02 (Military: 2021 - 2022)
Bachelor of Engineering, Computer Science GPA: 3.8/4.5

AWARDS & HONORS

- **ACM CHI Student Design Competition Finalist (Top 4)** 2025.05
First Author and Team Leader, ACM CHI Conference on Human Factors in Computing Systems
- **Ministry of Science and ICT 2024 XR Device Contents Makerthon Excellence Award** 2024.12
President's Award, National IT Industry Promotion Agency (NIPA)

PROFESSIONAL SERVICES

- **Reviewer**, IEEE Conference on Virtual Reality and 3D User Interfaces 2026

PUBLICATIONS

- CrossGaussian: Enhancing Remote Collaboration through 3D Gaussian Splatting and Real-time 360° Streaming – **Jaehyun Byun**, Byunghoon Kang, Yonghyun Gwon, Hongsong Choi, Yunseo Do, Eunho Kim, Sangkeun Park, and Seungjae Oh *UIST Adjunct 2025* (ACM Symposium on User Interface Software and Technology)
- Le Petit Care: A Child-Attuned Design for Personalized ADHD Symptom Management Through AI-powered Extended Reality – **Jaehyun Byun**, Chowon Joung, Yerim Lee, Suyun Lee, and Wooky Won *CHI EA 2025* (Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems)
- Metaverse Platform Specialized in Discussion and Debate Functions in Online Communication Systems – Minsu Kim, **Jaehyun Byun**, and Tack Woo *Journal of Digital Contents Society*, Vol. 25, No. 6, Jun. 2024

TECHNICAL SKILLS

Languages: C++, C#, Python, HLSL, JavaScript
Prototyping Tools: Unity, Arduino IDE, Raspberry Pi Pico, TensorFlow, Hugging Face
Design Tools: Fusion360, Figma, Blender, Miro
Statistics: ANOVA, Friedman test, Wilcoxon signed-rank test, R, SPSS
Certifications: SQL Developer (Korea Data Agency), AI Prompt Engineering Professional (Korea Productivity Center)

SCHOLARSHIPS

- **Donggyo Youth Prize, Award for Outstanding Undergraduate Achievements** Soorim Foundation

PROJECTS

- **CrossGaussian**, ITEM Lab (Kyung Hee University) 2024.09 - 2025.10
Real-Time Remote Collaboration Interaction Research Based on Camera Based Room-Scale Neural Rendering
 - Tools & Technologies: Azure DevOps, C++, C#, Python, HLSL, Unity, Photon, CUDA, FFmpeg
 - Role & Team: Research Lead | Client ×2, Backend ×2, Data Analysis ×1
 - Details: Addressed information asymmetry in remote collaboration by integrating room-scale 3DGS reconstruction with 360° video streaming; designed visualization and interaction design space through prior literature review; automated data collection and remote reconstruction pipeline by bridging Unity/HLSL client with Python/CUDA backend via SSH/SFTP; implemented rendering-based interaction techniques and validated hypotheses through N=24 user study with statistical analysis; accepted as poster at ACM UIST 2025 and Demo at IEEE ISMAR 2025
- **i-There**, P&C Solution 2024.06 - 2025.05
AR fairy-tale app with AI-driven real-time ADHD symptom detection and personalized behavioral intervention
 - Tools: Jira, C#, Python, Unity, MariaDB, TensorFlow, Blender, Figma
 - Role & Team: PM & Lead Developer | Planning ×2, Design ×3, Programming ×2, Medical AI ×1
 - Details: Structured 18 DSM-5 diagnostic criteria into digital behavioral metrics and designed user flow and art direction targeting children aged 8–12; implemented child-facing play content, built medical CNN model and deployed on-device via ONNX conversion; preprocessed and normalized user behavior and play data for AI consumption and built centralized DB-based log collection and processing pipeline; developed on P&C Solution's industrial-grade METALENSE2 AR device and won Excellence Award at MSIT XR Device Contents Makeathon 2024